

Project 3 — Optimal Risk-Averse Contracting Strategy for a Portfolio of Renewables

ECON-138

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Abstract

This document is the main teaching material for Project 3 of ECON-138 / ECON-238. Its purpose is to give the student part of the background needed to carry out the project tasks: the basics of forward contracting for electricity, risk-averse decision-making under uncertainty, and renewable portfolio selection under bilateral supply-side contracts. In this document, we also introduce the initial setup for the optimal risk-averse contracting strategy problem that will be the focus of Project 3.

1 Introduction

What a forward contract is

A *forward contract* on energy is a bilateral financial agreement between two parties — typically a generator (the *seller* or short position) and a consumer or trader (the *buyer* or long position) — that fixes a price P (\$/MWh) and a quantity Q (MWh) to be cleared against the realized *spot price* π of energy in the relevant delivery period. The contract is settled *financially*: no physical energy changes hands through the instrument itself. For each delivery period, the seller receives from the buyer the financial flow

$$R = (P - \pi) Q, \tag{1}$$

which is positive whenever $\pi < P$ (the realized spot is below the locked-in price, so the buyer compensates the seller) and negative whenever $\pi > P$ (the spot is above the contract price, and the seller compensates the buyer). The buyer's flow is the mirror image, $-(P - \pi) Q = (\pi - P) Q$, an analogous expression to (1) with the opposite sign, representing a net purchase cost. Physical energy is delivered separately, through the short-term spot market, at whichever price the market clears: generators continue to earn the spot revenue $\pi \cdot g$ on their physical output, and consumers continue to incur the spot cost $\pi \cdot d$ on their physical consumption. The interaction between the financial settlement of the forward contract and the generator's stochastic spot-market revenue is the central topic of this project.

Why the contract price departs from the expected spot

In practice, P and $\mathbb{E}[\pi]$ differ, and the gap reveals the *risk premia* each side is willing to pay.

- **Generators (sellers) accept a P below $\mathbb{E}[\pi]$** because they fear long stretches of low spot prices. In renewable-heavy systems this is not a tail event, it is the recurring pattern: hydro-dominated systems face wet-year cascades of low marginal water values, and wind and solar build-outs

produce windows in which zero-marginal-cost output depresses the whole system’s spot price. Facing a distribution with a persistent left tail, a risk-averse generator prefers a lower but *certain* P to a higher-mean, left-skewed spot stream.

- **Consumers (buyers) accept a P above $\mathbb{E}[\pi]$** because they fear short bursts of price spikes driven by droughts, thermal fuel shocks, unit outages, or transmission congestion. A fixed P caps their exposure to the right tail.

Both sides are trading *tail risk for certainty*. The forward contract is a *mutual hedge*: the seller gives up upside to cover the downside, the buyer gives up expected savings to cover the spikes. The difference $P - \mathbb{E}[\pi]$, the *forward risk premium*, measures the relative intensity of each side’s aversion to its own tail.

Virtues of forward contracting

Forward contracts are more than a risk-allocation device; they support several structural properties of modern power markets:

- *Revenue predictability.* The fixed-price leg PQ stabilizes the generator’s cash flow and tightens the profit distribution, a necessary condition for serviceable long-term debt.
- *Investment signal for new capacity.* Buyers willing to pay $P > \mathbb{E}[\pi]$ on long-horizon contracts signal scarcity and trigger greenfield investment. In Brazil, long-term forward contracts (Power Purchase Agreements, PPAs) have been the main financing vehicle for renewable build-out over the last two decades (Barbosa et al., 2024; Street et al., 2025).
- *Reduced spot-price volatility exposure.* By pre-committing volume, both sides cut their dependence on realized spot conditions and, in aggregate, reduce the welfare cost of spot-price variance.
- *Project financeability.* A generator holding long-term forward contracts can pledge those cash flows as collateral for project finance, unlocking capital costs that pure merchant exposure would not support.

Risks of forward contracting

The benefits come with costs. Three risk categories are standard, and a fourth one is specific to variable renewables:

- *Basis risk.* The reference price in the contract need not match the locational price the physical asset actually sees — e.g., a generator in the Brazilian northeast (NE) zone¹ who writes a contract indexed to the southeast (SE) hub sees a wedge whenever transmission between the two zones binds.
- *Counterparty risk.* Either side may default on the financial settlement, especially over long PPA horizons.
- *Inflexibility risk.* A large forward position taken at t_0 can become uneconomic under later state changes (technology prices, regulation, load); the contract is then renegotiated or bought back, usually at a loss.

¹A zone is a subsystem or node of the network where a zonal marginal price is defined.

The fourth risk is what motivates this project:

Definition 1.1 (Price-and-Quantity Risk (Barbosa et al., 2024)). *Consider a renewable generator committed to a long-term forward contract of volume Q at price P , with stochastic generation profile $\{g_t\}_{t \in T}$ and stochastic spot price $\{\pi_t\}_{t \in T}$. The Price-and-Quantity Risk (PQ-Risk) is the compounded financial exposure that materializes in two specific joint realizations of (π_t, g_t) :*

- (i) *a deficit in production relative to the contracted amount ($g_t < Q$) combined with a high spot price (π_t large), in which case the generator must clear the deficit at an elevated price;*
- (ii) *a surplus in production ($g_t > Q$) combined with a low spot price (π_t small), in which case the surplus generation is cleared at a price that fails to compensate its financial cost.*

Equivalently, the PQ-Risk is a two-factor, double-sided risk whose materialization requires the adverse realization of both the price factor and the quantity factor simultaneously, and cannot be hedged independently on each dimension.

PQ-Risk is not a tail event to be weathered. For wind and solar in renewable-heavy systems it is the recurring state of the world: the windy or sunny hours are precisely the ones in which zero-marginal-cost output floods the market and drags the spot price down; the calm or cloudy hours are the ones in which residual thermal capacity sets the price high. **The negative correlation between g_t and π_t that defines PQ-Risk is a structural feature of the system, not a statistical artifact.** Systems abundant in inflexible and seasonal resources tend to exhibit spot-price profiles that are negatively correlated with those resources.

Forward contracts stabilize revenue, but the joint randomness of price and renewable generation reintroduces a compounded risk that any serious contracting strategy has to price. That is the subject of this project.

2 Forward Contracts and the Profit of a Renewable Generator

2.1 Mathematical framework

We model a delivery horizon $T = \{1, 2, \dots, |T|\}$ (hours, days, or months, depending on granularity). All random quantities are defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$; a point $\omega \in \Omega$ is a *scenario*. Scenario realizations are written as $\pi_t(\omega)$, $g_t(\omega)$, and so on; we suppress ω whenever the context is clear.

Assumption 2.1 (Primitives).

- $P^{\text{sell}} \in \mathbb{R}_+$: fixed contract price negotiated with the consumer, in $\$/MWh$.
- $Q \in \mathbb{R}_+$: contracted quantity per period, in MWh/h (or $avgMW$).
- $\pi_t : \Omega \rightarrow \mathbb{R}$: random spot price at period t , in $\$/MWh$.
- $g_t : \Omega \rightarrow \mathbb{R}_+$: random generation of the asset at period t , in MWh .
- $c \in \mathbb{R}_+$: marginal cost of production. For renewables we set $c = 0$.

2.2 Revenue from the forward contract

The forward settlement is purely financial. For a single-period contract the seller's revenue is given by equation (1). Summing the net revenue over the contract horizon:

$$R = \sum_{t \in T} (P - \pi_t) Q. \quad (2)$$

This quantity is a random variable because $\{\pi_t\}_{t \in T}$ is random. The expected revenue in a given period t is $(P - \mathbb{E}[\pi_t])Q$, which collapses to zero if $P = \mathbb{E}[\pi_t]$ and is positive for the seller whenever $P > \mathbb{E}[\pi_t]$.

2.3 Total profit of a renewable generator holding a short forward position

The generator operates its plant independently of the contract: every MWh produced is sold into the spot market at the realized price $\pi_t(\omega)$, earning $\pi_t(\omega) g_t(\omega)$, and the financial settlement is added on top. For a single period and $c = 0$:

$$\Pi = (P - \pi) Q + \pi g. \quad (3)$$

For the multi-period case, with each period t taken to be one hour:

$$\Pi = \sum_{t \in T} [(P - \pi_t) Q + \pi_t g_t]. \quad (4)$$

A short algebraic rearrangement isolates the deterministic and stochastic components:

$$\Pi = P \cdot Q \cdot |T| + \sum_{t \in T} \pi_t (g_t - Q). \quad (5)$$

Equation (5) admits a direct economic reading. The first term, $P \cdot Q \cdot |T|$, is the deterministic revenue locked in by the forward contract: a fixed payment that does not depend on either price or generation. The second term, $\sum_{t \in T} \pi_t (g_t - Q)$, is the residual exposure to the spot market: at each period the generator clears the difference between its actual production and the contracted volume at the prevailing spot price. When $g_t > Q$, the surplus $g_t - Q$ is sold at π_t ; when $g_t < Q$, the deficit $Q - g_t$ has to be purchased at π_t to honour the contract.

This decomposition reveals what a forward contract does and does not do for the generator. If production were perfectly known and equal to the contracted volume in every period — that is, if $g_t = Q$ for all t — the residual term would vanish identically and the generator would face no spot-market uncertainty at all. Renewables, however, are precisely the case in which g_t is random, so the residual cannot be eliminated. The forward contract therefore creates an incentive to balance two opposing flows: the revenue from selling generation surpluses in the spot market in periods of overproduction, and the cost of buying generation deficits in the spot market in periods of underproduction. The optimal contracting decision therefore depends on two ingredients: the joint distribution of (π_t, g_t) , which governs how the second term realizes across scenarios, and the generator's risk-aversion profile, which sets the trade-off between the deterministic revenue of the first term and the stochastic exposure of the second. As Section 2.4 makes concrete, the negative correlation between price and renewable generation that defines PQ-Risk is what makes this trade-off non-trivial.

2.4 A single-period illustration of PQ-Risk

Consider a solar generator with $P = 50$, $Q = 10$, $c = 0$, and a single-period horizon $T = \{1\}$. Two equiprobable scenarios capture the negative correlation between renewable generation and spot price — low price paired with high generation, high price paired with low generation:

Scenario	π_1	g_1
ω_1 (sunny)	0	15
ω_2 (cloudy)	100	5

We can see the impact of the negative correlation between spot price and generation on the per-period spot revenue: the expected price is $\mathbb{E}[\pi] = 50$ and the expected generation is $\mathbb{E}[g] = 10$, but the expected spot revenue (or the market value) is $\mathbb{E}[\pi g] = (0 \times 15 + 100 \times 5) \times 1/2 = \250 — well below \$500, the result we would find if these two random variables were independent.

In addition, if we take $Q = \mathbb{E}[g] = 10$, the generator has nominally eliminated any volumetric mismatch on average; the two realized profits are nevertheless

$$\begin{aligned}\Pi(\omega_1) &= P \cdot Q + \pi(\omega_1)(g(\omega_1) - Q) = 500 + 0 \cdot 5 = 500, \\ \Pi(\omega_2) &= P \cdot Q + \pi(\omega_2)(g(\omega_2) - Q) = 500 + 100 \cdot (-5) = 0,\end{aligned}$$

In the high-generation scenario, the generator earns \$0 from clearing +5 MWh (+50% of its expected generation) in the spot market. However, in the low-generation scenario, clearing −5 MWh (−50% of its expected generation) in the spot market results in an expenditure equal to 100% of its contract revenue, preventing the generator from covering its fixed costs.

The reason is structural: The forward contract locks in a price, but the joint realization of (π, g) places more weight on the 5 MWh generation deficit (relative to the contract amount) than on the 5 MWh generation surplus. This is only one of the effects that impact contracting decisions. Another, not covered in this example but still relevant, is the asymmetry and kurtosis of the spot price (i.e., a heavy right tail).

So, contracting the expected quantity removes the expected volumetric mismatch and nothing more. By plotting the generator's profit as a function of the contracting amount Q for the two scenarios (Figure 1), it is clear that in the sunny scenario the best decision is to contract the full capacity (say, 15 MWh), whereas in the cloudy scenario the best decision is not to contract at all. Interestingly, for every Q between 0 and 15 MWh, the profit is a random variable with a constant expected value (equal to the expected spot revenue, \$250). This happens because $P = \mathbb{E}[\pi]$, which reduces the first term of the rearranged profit formula below to zero in expectation.

$$\Pi = (P - \pi)Q + \pi g$$

When analyzing worst-case scenarios, the picture is completely different. The worst-case outcome, which occurs with probability 50%, is that contracting at $Q = 0$ or $Q = \mathbb{E}[g] = 10$ may prevent the generator from covering fixed costs of, say, \$200.

It is worth noting, however, that there exists an interesting contract level ($Q^* = 5$ MWh) that maximizes the worst-case profit function and makes the random profit deterministic (both scenarios yield the same profit), i.e., not subject to uncertainty in either the market or generation. This is a special case that typically arises only in a two-scenario setting, but it reveals the essence of risk control. With more scenarios, the variance-reduction effect should still be observed for values of $Q < \mathbb{E}[g]$.

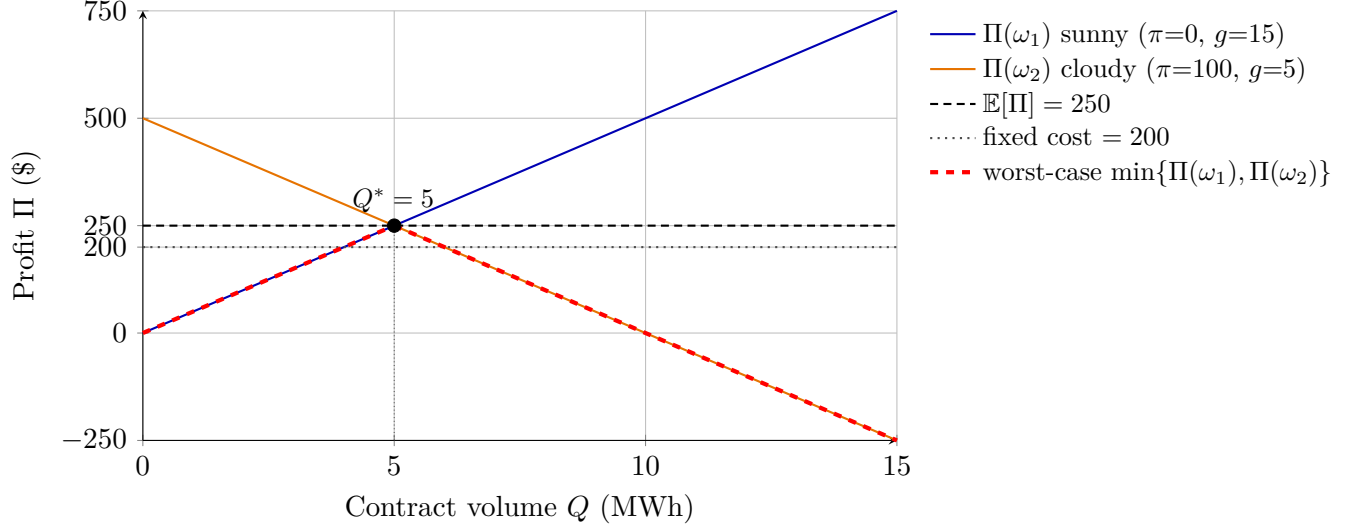


Figure 1: Profit Π as a function of the contract volume Q for the two equiprobable scenarios of the single-period illustration ($P = 50$, $\mathbb{E}[g] = 10$). The scenario lines are $\Pi(\omega_1) = 50Q$ (sunny) and $\Pi(\omega_2) = 500 - 50Q$ (cloudy). Their equiprobable average is the flat $\mathbb{E}[\Pi] = 250$, independent of Q because $P = \mathbb{E}[\pi]$. The worst-case profit at each Q is the *lower* of the two lines: it collapses to \$0 at $Q = 0$ (sunny) and at $Q = \mathbb{E}[g] = 10$ (cloudy), below the fixed-cost reference of \$200, and is maximized at $Q^* = 5$, where the two lines cross and the random profit becomes deterministic at \$250.

Therefore, in this setting, a risk-averse generator is likely to select $Q^* < \mathbb{E}[g]$ to secure profit in all scenarios rather than being exposed to the possibility of bankruptcy. This example is purely illustrative and, of course, does not capture all the nuances and possibilities of a forward and spot market strategy. Nevertheless, it highlights that a forward contract serves as a hedging tool but must be carefully optimized in accordance with the generator's risk-aversion profile.

3 Risk Aversion and the Certainty Equivalent

3.1 Preference functionals and the certainty equivalent

A *preference functional* is a mapping $\rho : \mathcal{X} \rightarrow \mathbb{R}$ from a space $\mathcal{X}$ of random variables (random profits, here) to the real line: a scalar summary of how the decision-maker evaluates a distribution. If ρ is such that $\rho(\beta) = \beta$ whenever β is a degenerate distribution (a deterministic number), then it is also the *certainty equivalent*, the sure amount judged indifferent to holding the random Π . A *risk-neutral* agent uses $\rho(\Pi) = \mathbb{E}[\Pi]$. A *risk-averse* agent uses a functional ρ such that, for any non-degenerate Π ,

$$\rho(\Pi) < \mathbb{E}[\Pi]. \quad (6)$$

The gap $\mathbb{E}[\Pi] - \rho(\Pi) \geq 0$ is the *risk premium*: the expected profit the decision-maker trades away to remove the randomness.

3.2 Conditional Value-at-Risk as a preference functional

In modern finance, the workhorse functional for risk-averse decisions over random profits is the *Conditional Value-at-Risk* (CVaR). CVaR_α at level $\alpha \in (0, 1)$ is the average outcome in the worst

α -fraction of scenarios. It differs from Value-at-Risk (VaR_α), which reports only the *threshold* of that worst fraction: VaR answers “where is the cutoff?”; CVaR answers “given we are in the bad zone, how bad is it on average?”. CVaR is sensitive to the severity of the tail, not only to its location.

Sign convention. CVaR is classically defined on *losses* (costs, negative returns). Because this project works with random *profits* Π , we flip the sign convention and operate on the *lower tail* of Π : the worst scenarios are those with the lowest profits. Throughout the project we use

$$\text{CVaR}_\alpha(\Pi) := \mathbb{E}[\Pi \mid \Pi \leq F_\Pi^{-1}(\alpha)], \quad (7)$$

where $F_\Pi^{-1}(\alpha)$ is the α -quantile of Π (the level below which the profit falls with probability α), and α is the *tail fraction* — e.g., $\alpha = 0.05$ means “the average profit in the worst 5% of scenarios”. A decision-maker maximizing $\text{CVaR}_\alpha(\Pi)$ is focused on protecting the lower tail of profit.²

3.3 The combined risk metric

Decision-makers rarely maximize CVaR alone: focusing only on the worst tail throws away expected performance. The standard compromise is a convex combination of the expected value and CVaR:

Definition 3.1 (Certainty-equivalent risk metric). *For a random profit Π with $\mathbb{E}[|\Pi|] < \infty$ and risk-aversion parameters $\alpha \in (0, 1)$ and $\lambda \in [0, 1]$,*

$$\rho_{\alpha,\lambda}(\Pi) := \lambda \text{CVaR}_\alpha(\Pi) + (1 - \lambda) \mathbb{E}[\Pi]. \quad (8)$$

The convex-combination form (8) has an interpretation as a probability-dependent utility function; see Street (2010).

Both α and λ are risk-aversion parameters, but they act on different axes. The tail fraction $\alpha \in (0, 1)$ sets *which* part of the profit distribution the decision-maker focuses on; in practice α is small — typically $\alpha = 0.05$ or $\alpha = 0.01$, so that $\text{CVaR}_\alpha(\Pi)$ reports the average profit in the worst 5% or 1% of scenarios. The mixing weight $\lambda \in [0, 1]$ then sets *how much* that worst-case value counts relative to the expected profit in the final certainty-equivalent metric: $\lambda = 0$ is risk-neutral, $\lambda = 1$ is fully tail-focused, and intermediate values interpolate between the two. Together, (α, λ) jointly encode the sharpness of the tail focus and its weight in the overall metric.

3.4 Two equivalent formulations of CVaR

Equation (7) is the *conditional* formulation: good for intuition, bad for optimization (the conditioning event depends on the distribution of Π , which itself depends on the decision variables). Rockafellar and Uryasev (2000) recast CVaR_α as the optimal value of a concave maximization problem, unlocking the LP form:

Proposition 3.2 (Rockafellar–Uryasev representation, lower-tail profit version). *For a random profit $\Pi : \Omega \rightarrow \mathbb{R}$ with $\mathbb{E}[|\Pi|] < \infty$ and $\alpha \in (0, 1)$,*

$$\text{CVaR}_\alpha(\Pi) = \max_{z \in \mathbb{R}} H_\alpha(\Pi, z), \quad H_\alpha(\Pi, z) := z - \alpha^{-1} \mathbb{E}[(z - \Pi)^+], \quad (9)$$

where $(x)^+ := \max(x, 0)$. Letting $F_\Pi(z) := \mathbb{P}(\Pi \leq z)$ denote the cumulative distribution function of Π , the smallest optimal z^* is the α -quantile of Π :

$$z^* = F_\Pi^{-1}(\alpha) := \inf\{z \in \mathbb{R} : F_\Pi(z) \geq \alpha\}. \quad (10)$$

The proof is a first-order optimality argument, worked out in Section 3.5 below.

²Some references use α as the confidence level ($\alpha = 0.95$) rather than the tail fraction ($\alpha = 0.05$). Same object, different convention. We use the tail-fraction convention throughout; it lines up with the Rockafellar–Uryasev formulation below.

3.5 Derivation of $z^* = F_{\Pi}^{-1}(\alpha)$

We derive Proposition 3.2 from scratch by a chain of differentiations, working directly with the profit Π to match the notation of the rest of the project.

Assumption 3.3 (Continuity at the quantile). *The distribution of Π is continuous at z^* : $\mathbb{P}(\Pi = z^*) = 0$. Under this assumption the α -quantile $F_{\Pi}^{-1}(\alpha)$ is a single point and $\mathbb{P}(\Pi < z^*) = \mathbb{P}(\Pi \leq z^*) = \alpha$.*

Assumption 3.3 is harmless for continuous joint distributions of (π_t, g_t) . In the sampled LP (19) below, F_{Π} has atoms and z^* need not be unique; the Rockafellar–Uryasev LP still returns the correct CVaR_{α} by picking *any* optimal z^* from the quantile set.

Let $f(\Pi, z) := H_{\alpha}(\Pi, z) = z - \alpha^{-1} \mathbb{E}[(z - \Pi)^+]$. The function f is concave in z (it is the sum of a linear term and the negative of a convex function; $(\cdot)^+$ is convex, and $\mathbb{E}$ preserves convexity), so the first-order condition (FOC) — the requirement that $\partial f / \partial z = 0$ at an interior optimum — is both necessary and sufficient for a maximum.

Step 1. Differentiate f in z . Exchange of differentiation and expectation is justified by the boundedness of $|(z - \Pi)^+|$ on any compact z -interval and the integrability of Π :

$$\frac{\partial f(\Pi, z)}{\partial z} = 1 - \alpha^{-1} \mathbb{E} \left[\frac{\partial}{\partial z} (z - \Pi)^+ \right]. \quad (11)$$

Step 2. Differentiate the positive part. The map $x \mapsto (x)^+$ has the derivative

$$\frac{d}{dx} (x)^+ = \mathbf{1}_{\{x > 0\}}(x), \quad (12)$$

which is the indicator of the strict positivity set. (At $x = 0$ the map is not differentiable in the classical sense, but any measurable selection of the subdifferential works; the resulting FOC is independent of the choice on a null set when the distribution of Π has no atom at z .)

Plugging (12) into (11) with $x = z - \Pi(\omega)$:

$$\begin{aligned} \frac{\partial f(\Pi, z)}{\partial z} &= 1 - \alpha^{-1} \mathbb{E} \left[\mathbf{1}_{\{z - \Pi(\omega) > 0\}} \cdot 1 \right] \\ &= 1 - \alpha^{-1} \mathbb{E} \left[\mathbf{1}_{\{\Pi(\omega) < z\}} \right]. \end{aligned} \quad (13)$$

Step 3. Identify the probability. By definition of the cumulative distribution function,

$$\mathbb{E} \left[\mathbf{1}_{\{\Pi(\omega) \leq z\}} \right] = \mathbb{P}(\Pi \leq z) = F_{\Pi}(z), \quad (14)$$

and when the distribution of Π is continuous at z , the difference between $\{\Pi(\omega) < z\}$ and $\{\Pi(\omega) \leq z\}$ has zero probability. Substituting (14) into (13):

$$\frac{\partial f(\Pi, z)}{\partial z} = 1 - \alpha^{-1} F_{\Pi}(z). \quad (15)$$

Step 4. Set the derivative to zero. The first-order condition for an interior maximum is

$$1 - \alpha^{-1} F_{\Pi}(z^*) = 0, \quad (16)$$

which rearranges to

$$\boxed{F_{\Pi}(z^*) = \alpha, \quad \text{i.e.,} \quad z^* = F_{\Pi}^{-1}(\alpha),} \quad (17)$$

the α -quantile of Π , which is exactly the Value-at-Risk in the *profit* convention used here (lower-tail VaR on profits at tail fraction α): $z^* = \text{VaR}_{\alpha}(\Pi)$.

3.6 Consequences of the derivation

Two facts follow immediately from (17).

- *Equivalence of the two formulations.* Under Assumption 3.3, $\mathbb{P}(\Pi \leq z^*) = \alpha$, so substituting $z^* = F_\Pi^{-1}(\alpha)$ into (9) and recognizing $\mathbb{E}[(z^* - \Pi)^+] = (z^* - \mathbb{E}[\Pi \mid \Pi \leq z^*]) \mathbb{P}(\Pi \leq z^*) = (z^* - \mathbb{E}[\Pi \mid \Pi \leq z^*]) \alpha$ gives

$$\text{CVaR}_\alpha(\Pi) = z^* - \alpha^{-1}(z^* - \mathbb{E}[\Pi \mid \Pi \leq z^*])\alpha = \mathbb{E}[\Pi \mid \Pi \leq F_\Pi^{-1}(\alpha)],$$

recovering the conditional formulation (7).

- *Measure-theoretic and sampled forms.* Equation (9) can be rewritten in a form that makes its linear-programming structure transparent. Introducing a non-negative slack function $\delta : \Omega \rightarrow \mathbb{R}_+$ to absorb the positive part $(z - \Pi)^+$, the unconditional representation becomes

$$\text{CVaR}_\alpha(\Pi) = \max_{z \in \mathbb{R}, \delta(\cdot) \geq 0} \left\{ z - \alpha^{-1} \int_\Omega \delta(\omega) dP(\omega) : \delta(\omega) \geq z - \Pi(\omega), \text{ } P\text{-a.e. } \omega \in \Omega \right\}, \quad (18)$$

a (generally infinite-dimensional) convex program over the scalar z and the function $\delta(\cdot)$. When the true sample space Ω is continuous or intractably large, we work instead with a sample $\Omega_N \subset \Omega$ of N equiprobable scenarios drawn from (Ω, P) . The integral collapses to a finite sum and the program becomes the linear program

$$\widehat{\text{CVaR}}_\alpha(\Pi) = \max_{z, \{\delta_\omega\}_{\omega \in \Omega_N}} \left\{ z - \frac{1}{\alpha N} \sum_{\omega \in \Omega_N} \delta_\omega : \delta_\omega \geq z - \Pi(\omega), \delta_\omega \geq 0, \forall \omega \in \Omega_N \right\}, \quad (19)$$

whose optimum $\widehat{\text{CVaR}}_\alpha(\Pi)$ is a sample-average approximation of the true $\text{CVaR}_\alpha(\Pi)$, consistent as $N \rightarrow \infty$. This sampled form is the structural reason why risk-averse contracting problems are tractable at scale whenever $\Pi(\omega)$ is linear in the decision variables.

3.7 Coherent-risk properties of $\rho_{\alpha,\lambda}$

The certainty-equivalent metric $\rho_{\alpha,\lambda}$ is a *coherent risk measure* on profits, in the sense of Artzner et al. (1999) adapted from the original loss convention by a sign flip. This follows from the coherence of CVaR_α on the lower tail of profits combined with the linearity of $\mathbb{E}[\cdot]$: both satisfy the five axioms below, and the convex combination preserves each of them.

Properties of $\rho_{\alpha,\lambda}$. For random profits Π, Π_1, Π_2 and real $\beta \geq 0, c \in \mathbb{R}$:

- *Superadditivity (diversification).* $\rho_{\alpha,\lambda}(\Pi_1 + \Pi_2) \geq \rho_{\alpha,\lambda}(\Pi_1) + \rho_{\alpha,\lambda}(\Pi_2)$. Pooling two profit streams produces a tail-adjusted value no worse than the sum of the individual tail-adjusted values; combining assets whose bad scenarios do not coincide improves the aggregate metric.
- *Positive homogeneity.* $\rho_{\alpha,\lambda}(\beta \Pi) = \beta \rho_{\alpha,\lambda}(\Pi)$ for $\beta \geq 0$. Scaling the business scales the metric by the same factor.
- *Translation equivariance (shift-additivity).* $\rho_{\alpha,\lambda}(\Pi + c) = \rho_{\alpha,\lambda}(\Pi) + c$ for deterministic c . A sure cash addition increases the metric by exactly that amount.
- *Monotonicity.* If $\Pi_1 \geq \Pi_2$ almost surely, $\rho_{\alpha,\lambda}(\Pi_1) \geq \rho_{\alpha,\lambda}(\Pi_2)$. A stochastically dominating profit is preferred.

(v) *Concavity in Π* . For any Π_1, Π_2 and $\theta \in [0, 1]$,

$$\rho_{\alpha,\lambda}(\theta \Pi_1 + (1 - \theta) \Pi_2) \geq \theta \rho_{\alpha,\lambda}(\Pi_1) + (1 - \theta) \rho_{\alpha,\lambda}(\Pi_2). \quad (20)$$

This is a direct consequence of properties (i) and (ii): superadditivity gives $\rho_{\alpha,\lambda}(\theta \Pi_1 + (1 - \theta) \Pi_2) \geq \rho_{\alpha,\lambda}(\theta \Pi_1) + \rho_{\alpha,\lambda}((1 - \theta) \Pi_2)$, and positive homogeneity then collapses each term to $\theta \rho_{\alpha,\lambda}(\Pi_1)$ and $(1 - \theta) \rho_{\alpha,\lambda}(\Pi_2)$.

3.8 Why coherence matters for optimization

Each of the five properties above does concrete work when $\rho_{\alpha,\lambda}$ is placed as the objective of an optimization problem:

- *Concavity of $\rho_{\alpha,\lambda}$ in Π* , combined with linearity of Π in the decisions (forward volumes, purchase quantities), turns the maximization problem in Section 4 into a *convex program* — a linear program when the profit is affine and the scenario set is finite. No spurious local optima to chase.
- *Superadditivity* is why diversification is mechanically rewarded. For a vector of weights across assets with imperfectly correlated joint distributions, the objective strictly improves with a spread allocation over any concentrated one, *at no cost in expected profit*. This is the mathematical statement of the complementarity intuition in the Brazilian wind-biomass-small-hydro portfolio.
- *Positive homogeneity* means the scale of the decision is a free parameter: the optimal portfolio shape is independent of its size. The contract-structure problem decouples from the firm-size problem.
- *Translation equivariance* lets the decision-maker separate stochastic from deterministic components of profit: a fixed-fee payment enters the objective additively and does not have to be scenario-traced.

These are not abstract conveniences. They are the properties that make risk-averse stochastic programming solvable at the operational scale of an energy portfolio (Shapiro et al., 2021).

4 Optimal Contracting and Portfolio Management

We now assemble the pieces from Sections 2–3 into the full problem: a Trading Company (TC) builds a portfolio of renewable assets through bilateral supply-side contracts and simultaneously sells a forward contract to a consumer.

4.1 Setting

We model a price-taking Trading Company (TC) operating over a delivery horizon $T = \{1, 2, \dots, |T|\}$, with one hour per period. Following the sampled formulation of Section 3, randomness in prices and generation is represented by a sample $\Omega_N \subset \Omega$ of N equiprobable scenarios, where each $\omega \in \Omega_N$ is a joint realization of all random variables across the horizon.

Generation portfolio. The TC faces a menu of candidate generators $i \in \mathcal{I} = \{\text{Bio}, \text{SH}, \text{Wind}\}$, each located in a zone $n(i) \in \{\text{NE}, \text{SE}\}$:

Asset i	Zone $n(i)$	Energy Certificate EC^i (MWh/h)	Nature of generation
Bio (biomass)	SE (southeast)	17.59	deterministic seasonal profile
SH (small hydro)	SE (southeast)	17.00	stochastic (inflow-driven)
Wind (wind)	NE (northeast)	11.41	stochastic (weather-driven)

For each unit i , $g_{t\omega}^i$ denotes the random generation at period t in scenario ω (in MWh), and EC^i is the unit's *Energy Certificate* — the long-run average generation listed in the table. The dimensionless ratio

$$\text{GSF}_{t\omega}^i := \frac{g_{t\omega}^i}{\text{EC}^i} \quad (21)$$

is the *Generation Scaling Factor* of unit i : a normalized index whose unconditional mean is unity by construction. It is the natural way to express what fraction of the unit's long-run output is realized at (t, ω) , and it will let us write both contract-format profits below in a uniform way. Marginal costs are set to zero for all units, $c^i = 0$, consistent with the renewable nature of the portfolio.

Spot prices. The spot price in zone $n \in \{\text{NE}, \text{SE}\}$ at period t in scenario ω is denoted $\pi_{t\omega}^n$ (in \$/MWh). The joint distribution of $\{\pi_{t\omega}^{\text{NE}}, \pi_{t\omega}^{\text{SE}}, g_{t\omega}^i\}$ over $T \times \Omega_N$ is a primitive of the model.

Consumer-side contract. The TC also faces a free consumer in the NE zone willing to take a forward contract at a fixed price P^{sell} (\$/MWh) for a fixed quantity Q^{sell} per period (in MWh/h, equivalently avgMW). The volume sold forward to the consumer, $Q^{\text{sell}} \geq 0$, is a decision variable bounded above by a regulatory ceiling $Q_{\text{max}}^{\text{sell}}$. Because the consumer is located in NE, the forward contract clears against the NE zone price $\pi_{t\omega}^{\text{NE}}$ in every period; if the consumer were located in SE, π^{NE} would be replaced by π^{SE} .

Supply-side contract formats. The TC acquires generation from each asset i under one of two formats, chosen ex ante for that asset and fixed throughout. In both formats, $Q^i \in [0, \text{EC}^i]$ is the volume contracted from unit i (a decision variable), and P^i is the per-unit contract price (\$/MWh, a parameter set ex ante). The two formats differ in how the per-period cash flow on unit i is computed:

- **Tolling contract.** The TC pays a fixed fee $P^i Q^i$ per period (\$/h) to “rent” a fraction Q^i/EC^i of asset i 's capacity. It then *receives the generator's scaled production profile*: for a partial rental with $Q^i \leq \text{EC}^i$, the delivered generation at time t in scenario ω is

$$g_{t\omega}^i \cdot \frac{Q^i}{\text{EC}^i} = \text{GSF}_{t\omega}^i Q^i. \quad (22)$$

All the price and quantity risk on the supply side then sits with the TC: the generator collects the flat $P^i Q^i$ regardless of the realization of its own output.

- **Pay-as-produced (PAP) contract.** The TC pays a fixed unit price P^i per MWh of *actual* generation delivered, with the volume still scaled by the contracted share Q^i/EC^i . No fixed retainer is paid; the generator's upside and downside are only partially shifted to the TC, which makes PAP a *volumetrically softer* contract — and, as usual for such contracts, it commands a higher expected price than tolling.

4.2 Fundamental question

Decisions are evaluated through the certainty-equivalent metric $\rho_{\alpha,\lambda}$ developed in Section 3, parameterized by the tail fraction $\alpha \in (0, 1)$ and the mixing weight $\lambda \in [0, 1]$. The TC's problem can then be stated as follows: given the joint distribution of $\{\pi_{tw}^{\text{NE}}, \pi_{tw}^{\text{SE}}, g_{tw}^i\}$ and a risk-aversion profile (α, λ) ,

*How much forward volume Q^{sell} should the TC sell to the consumer,
and how much Q^i (and in what contract format) should it buy from each generator $i \in \mathcal{I}$,
so as to maximize the certainty equivalent of total profit?*

The next two subsections build the per-period profit Π_{tw}^{TC} under each contract format; the optimization problem is then assembled in Section 4.5.

4.3 Profit under the tolling format

Under tolling, for each unit i : (a) the TC pays a fixed retainer $P^i Q^i$ per period, and (b) the TC receives the scaled generation and sells it into the unit's zone at spot. Combined with the forward sale to the consumer, the TC's total profit in hour t under scenario ω is

$$\Pi_{tw}^{\text{TC,toll}} = (P^{\text{sell}} - \pi_{tw}^{\text{NE}}) Q^{\text{sell}} + \sum_{i \in \mathcal{I}} \left[-P^i Q^i + \pi_{tw}^{n(i)} \text{GSF}_{tw}^i Q^i \right] \quad (23)$$

for every $t \in T$, $\omega \in \Omega_N$. Three economic ingredients appear:

- forward-contract settlement against the consumer: $(P^{\text{sell}} - \pi_{tw}^{\text{NE}}) Q^{\text{sell}}$ — a function of the consumer-zone spot price only;
- fixed generation-contract retainer: $-\sum_i P^i Q^i$ — deterministic outflow;
- spot revenue of the scaled generation taken under tolling: $+\sum_i \pi_{tw}^{n(i)} \text{GSF}_{tw}^i Q^i$.

4.4 Profit under the pay-as-produced format

Under PAP, the TC pays P^i per delivered MWh (not a flat fee), and the delivered volume is still scaled by Q^i/EC^i because the contract is for a share. The per-period cash flow on unit i is then $-(P^i - \pi_{tw}^{n(i)}) \text{GSF}_{tw}^i Q^i$:

$$\Pi_{tw}^{\text{TC,PAP}} = (P^{\text{sell}} - \pi_{tw}^{\text{NE}}) Q^{\text{sell}} - \sum_{i \in \mathcal{I}} (P^i - \pi_{tw}^{n(i)}) \text{GSF}_{tw}^i Q^i \quad (24)$$

The structural difference between (23) and (24) is that under PAP the TC's exposure to unit i scales with *actual* generation GSF_{tw}^i , not with the contracted share Q^i/EC^i alone. High-generation hours in which $\pi_{tw}^{n(i)} > P^i$ produce positive TC cash flow; low-generation hours with $\pi_{tw}^{n(i)} < P^i$ hurt less because the contracted volume shrinks with the realized generation.

4.5 The optimization problem

The optimization problem set up below applies to either supply-side contract format introduced in Sections 4.3 and 4.4: we treat the per-period scenario profit $\Pi_{tw}^{\text{TC}}(\mathbf{q})$ as a generic random variable, given by (23) or (24) according to the format chosen ex ante for each supply-side asset, and we do not redefine it here.

All expectations and CVaR functionals below are computed as sample averages over the scenario set Ω_N introduced in Section 4.1; each $\omega \in \Omega_N$ carries probability $1/N$, and the Rockafellar–Uryasev linearization (19) produces a sample-average approximation $\hat{\rho}_{\alpha,\lambda}(\Pi^{\text{TC}})$ of the true certainty equivalent.

Horizon profit and the TC’s problem. The total scenario profit over the horizon is

$$\Pi_{\omega}^{\text{TC}}(\mathbf{q}) := \sum_{t \in T} \Pi_{t\omega}^{\text{TC}}(\mathbf{q}), \quad \omega \in \Omega_N, \quad (25)$$

where $\mathbf{q} := (Q^{\text{sell}}, Q^{\text{Bio}}, Q^{\text{SH}}, Q^{\text{Wind}}) \in \mathbb{R}_+^4$ is the decision vector. The TC’s problem is to choose $\mathbf{q}$ to maximize the certainty equivalent of total profit:

$$\max_{\mathbf{q}} \rho_{\alpha,\lambda}(\Pi^{\text{TC}}(\mathbf{q})). \quad (26)$$

Note that the certainty-equivalent objective is itself a function of the decision vector $\mathbf{q}$ through $\Pi^{\text{TC}}(\mathbf{q})$: each choice of $\mathbf{q}$ generates a different random profit, and $\rho_{\alpha,\lambda}$ scores it accordingly. Expanding $\rho_{\alpha,\lambda}(\Pi^{\text{TC}}(\mathbf{q})) = \lambda \text{CVaR}_{\alpha}(\Pi^{\text{TC}}(\mathbf{q})) + (1 - \lambda) \mathbb{E}[\Pi^{\text{TC}}(\mathbf{q})]$ and instantiating the Rockafellar–Uryasev sampled linearization (19) on Ω_N , the problem becomes a linear program in $(\mathbf{q}, z, \{\delta_{\omega}\}_{\omega \in \Omega_N})$:

$$\begin{aligned} \max_{\mathbf{q}, z, \{\delta_{\omega}\}} \quad & \lambda \left(z - \frac{1}{\alpha N} \sum_{\omega \in \Omega_N} \delta_{\omega} \right) + (1 - \lambda) \frac{1}{N} \sum_{\omega \in \Omega_N} \Pi_{\omega}^{\text{TC}}(\mathbf{q}) \\ \text{subject to} \quad & \delta_{\omega} \geq z - \Pi_{\omega}^{\text{TC}}(\mathbf{q}), & \forall \omega \in \Omega_N, \\ & \delta_{\omega} \geq 0, & \forall \omega \in \Omega_N, \\ & 0 \leq Q^{\text{sell}} \leq Q_{\max}^{\text{sell}}, \\ & 0 \leq Q^i \leq \text{EC}^i, & \forall i \in \mathcal{I}. \end{aligned} \quad (27)$$

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